

STATE-OF-THE-ART REVIEW

Sudden Cardiac Death in Athletes

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CME/MOC Objective for This Article: Upon completion of this activity, the learner should be able to: 1) describe the epidemiology of SCD in the

athlete population; 2) discuss the relevant elements of a pre-participation cardiovascular screening examination for risk stratification of competitive athletes; and 3) identify components of an emergency action plan for schools and organizations hosting athletic events or training facilities.

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ABSTRACT

Sudden cardiac death is a tragedy at any age and under any circumstances but is perhaps most tragic when it claims the life of the athlete, the individual who epitomizes health and a healthy lifestyle. Sports cardiologists from around the world have worked to quantitate the incidence of sudden cardiac death (SCD) in the athlete, to identify risk factors, to develop pre-participation screening tools, and to formulate plans to deal with on-field SCD. Progress has been made, but much remains to be done in order to make both competitive and recreational sports safer for both patients with known cardiac disease and athletes without known or suspected cardiac abnormalities. (J Am Coll Cardiol HF 2018;6:30-40) © 2018 Published by Elsevier on behalf of the American College of Cardiology Foundation.

Sudden cardiac death (SCD) in an apparently healthy athlete is a tragic event, counterintuitive to the common association between athletic activity and good health. These tragedies are highly publicized and generate considerable attention in the local community, which may then commit significant financial resources to prevent future events. The medical community, in turn, looks to the cardiac care team for guidance on the identification of athletes at risk, primary and secondary prevention strategies, and treatment.

Addressing this issue is complex as athletes come in all shapes, sizes, and ages. Sports can require various levels of cardiac performance, even within the same sport. The position played, the training required, and the environmental stresses make each athlete an individual study of risk and benefit. Data for athletes and SCD come from disparate sources, and expert consensus drives many published recommendations. This review informs the provider community of the current state of knowledge regarding the athletes at risk for SCD.

INCIDENCE OF SUDDEN CARDIAC DEATH

The true incidence of SCD in an athletic population is controversial. Estimates from studies in the United States and Europe are all complicated by various methodological issues ([Table 1](#)). The basis of these discrepancies revolves around what is included in the numerator (number of athletes who have experienced SCD) and the denominator (number of athletes at risk) of these equations. Some studies have included only events that resulted in death ([1-7](#)), whereas others also include those that survive sudden cardiac arrest (SCA) ([8-12](#)). Methods relying on media reports, passive registries, and catastrophic insurance claims have been used predominantly to

develop incidence reports which may underestimate SCD incidence. Drezner et al. ([11](#)) examined a population of Minnesota high school athletes over a 10-year period, using media reports from the Parent Heart Watch database and found 6 cases of SCD compared to only 1 case identified by using catastrophic insurance claims. The variability in reporting among National Collegiate Athletic Association (NCAA) athletes was demonstrated by Harmon et al. ([7](#)), who identified SCD from 2003 to 2013 through 3 different sources: 1) the NCAA Resolutions List; 2) the Parent Heart Watch database; and 3) NCAA insurance claims. Media database reports identified 70% of the total SCD cases, whereas insurance claims captured only 11% of cases.

Sudden cardiac death is a relatively infrequent event, and to study it, large, stable populations must be observed over a period of many years to account for random variability in death rates. Corrado et al. ([2](#)) published one of the first large scale studies of sudden cardiac death in athletes in the Veneto region of Italy. Over the final period of the study (2003 to 2004), the incidence of SCD was 0.4 per 100,000 person years ($n = 1$ death), whereas in the initial period (1979 to 1980), it was 3.6 per 100,000 person years ($n = 8$ deaths) with an average of approximately 2 deaths per year over the entire 26-year study period. This was similar to findings in an Israeli study by Steinvel et al. ([9](#)) over a 30-year time frame with a cumulative incidence of 2.6 per 100,000 person-years ($n = \sim 1$ death/year). However, there was a noted spike in deaths during their 30-year study period from 1995 to 1996, with an incidence of 8.4/100,000 person-years, whereas 2 periods (1989 to 1990 and 1993 to 1994) had zero deaths.

A consistent finding across studies is that male athletes have a 3 to 5 times higher incidence of SCD

ABBREVIATIONS AND ACRONYMS

ARVC = arrhythmogenic right
ventricular cardiomyopathy

DCM = dilated cardiomyopathy

HCM = hypertrophic
cardiomyopathy

SADS = sudden arrhythmic
death syndrome

SCA = sudden cardiac arrest

SCD = sudden cardiac death

than female athletes (1,13). Data from NCAA athletes also indicate that black athletes (incidence rate ratio of 3.2 compared to white athletes) and in particular Division I basketball players (1:5,200 athlete-years for all males and 1:4,380 athlete-years for black males) have a markedly higher incidence of SCD (7). Recognizing heterogeneity may help with efforts to target preventive strategies for groups with the highest risk.

The burden of SCD in athletic individuals must be considered in context of the risk of SCD in the general population. Studies indicate that the physical endeavor of sport activity and training poses a 2.4 to 4.5 increased risk of SCA and SCD relative to that in nonathletes and recreational athletes (2,10,14). Although sporting activity appears to increase the relative risk of SCD, the absolute number of cases is larger in the nonsporting population. Over the 10-year period of the Oregon Sudden Unexpected Death Study, there were 1,184 nonsport-associated SCA compared to 63 sport-associated SCA for patients 35 to 65 years of age (15). There are an estimated 201,286 cases of SCA/SCD across the general population in the United States annually (16), and, although extrapolation from the Oregon experience may not be entirely justified, the absolute number of SCD cases attributable to athletic activity will be a relatively small number when considering issues of population health.

PRIMARY PREVENTION AND SCREENING STRATEGIES

A pre-participation evaluation (PPE) prior to competing in athletics has been recommended by multiple societies (17–22) to identify or raise the suspicion of cardiovascular abnormalities that could potentially result in SCD on the athletic field (17). The appropriate components of the PPE are a subject of significant controversy and vary by country, league, and level of competition.

HISTORY AND PHYSICAL EXAMINATION. The American Heart Association (AHA) has published guidelines for the pre-participation screening of competitive athletes, which consist of a 14-point history and physical examination (23). The 14-point PPE is a Class I recommendation in the most recent Eligibility and Disqualification Recommendations for Competitive Athletes study published by the American College of Cardiology (ACC)/AHA (17). The Pre-Participation Physical Evaluation Monograph, 4th edition (PPE-4), is a joint project of 6 medical

organizations whose members are intimately involved in pre-participation assessments (21) and is endorsed by the AHA (23) (Table 2 compares the AHA 14 points with the PPE-4 elements). Despite the fact that these and similar recommendations have been in place for decades, studies indicate the general lack of awareness and incorporation into many state and school district policies in the United States (24–28).

The PPE medical history questions were developed through expert consensus and have not been scientifically validated in a prospective study. A systematic review/meta-analysis from Harmon et al. (7) reported a sensitivity and specificity for the detection of serious cardiac abnormalities to be 20% and 94%, respectively, for the history and 9% and 97%, respectively, for the physical examination. This likely reflects a high number of initial positive responses (24% to 68%) reported by high school and college athletes on questionnaires (29–31). Although initial responses on the PPE form are meant to be reviewed by a physician to determine if further evaluation is warranted, a study from Fudge et al. (31) suggested that 46% of initial positive responses necessitated further evaluation even after physician review. The AHA 14-point PPE and the PPE-4 monograph are considered the bases of any PPE and efforts to standardize them across institutions and states in the United States are crucial to their success.

ELECTROCARDIOGRAPHY. The addition of an electrocardiography (ECG) to the standard PPE has been mandated in Italy (2) and Israel (9) and has been recommended by the European Society of Cardiology (20) and International Olympic Committee (32). Data suggest the sensitivity of the ECG to detect underlying cardiovascular abnormalities that could place the athlete at risk of sudden cardiac death is superior to that of the history and physical examination (7). However, the addition of a resting ECG to screening protocols has not been evaluated in a prospective study. Data from Israel (9), which mandates ECG screening, and Minnesota (8), where ECG screening does not take place, suggest rates of SCD similar to that of Italy after the implementation of their mandated ECG screening program (9). In the United States, the issue of ECG screening in the collegiate athlete has been a topic of discussion in the lay press, although no mandate has emerged from the debate (22,33–35). Unfortunately, given the logistics and including financial costs and sheer volume of participants needed, a randomized, controlled trial of screening to prevent sudden cardiac death in the athlete is likely unrealistic.

TABLE 1 Studies of SCA and SCD Incidence in Athletic Populations

First Author (Ref. #)	Year	Incidence (per Athlete-Person Yrs)	Total Number of Cases	Definition(s)	Time Frame of Events	Population	Collection Method(s)	Age Range (yrs)	Study Period
Van Camp et al. (1)	1995	1/281,000	107	SCD	C/T	17 U.S. high school and college sports (excluded intramural, club, and PE classes)	Retrospective from National Center for Catastrophic Sports Injury Research database and media database	17–24	1983–1993
Corrado et al. (2)	2006	1/115,000 over the final 11 yrs (1/53,000 over the total period)	55	SCD	All	Veneto region, Italy	Prospective registry	12–35	1979–2004
Maron et al. (8)	2009	1/103,000	22	SCA+D	All	MN high school, collegiate, and professionals	Catastrophic Insurance Records; U.S. National Registry of Sudden Death in Athletes; NCAA website/communications; public records	12–31	1985–2007
Maron et al. (8)	2009	1/164,000	1,049	SCA+D	All	U.S. competitive organized team or individual sports (intramural/recreational excluded)	U.S. National Registry of Sudden Death in Athletes (media reports, directly submitted, NCCSIR, NHLBI)	8–39	1980–2006
Holst et al. (3)	2010	1/83,000	15	SCD	C/T	Competitive athletes	Denmark nationwide death certificates	15–35	2000–2006
Solberg et al. (4)	2010	1/117,000	23	SCD	All	During, or in connection with, physical activity	Retrospective Norwegian Cause of Death Registry, medical records, and autopsy reports from hospitals and clinics	15–34	1990–1997
Steinvil et al. (9)	2011	1/39,000	24	SCA+D	All	Israel Sport Law (“individuals who engage in sportive activity at any level of physical endurance”)	Media reports	12–44	1985–2009
Marijon et al. (10)	2011	1/102,000	50	SCA+D	C/T	Competitive athletes in France: participated in an organized sports program (team or individual) that required regular competition and training (excludes intramural sports)	Prospective population based using EMS responding to cardiac cases; media reports	10–35	2005–2010
Roberts and Stovitz (5)	2013	1/417,000	4	SCD	C/T	MN State High School League	Catastrophic Insurance Records	12–19	1993–2012
Drezner et al. (11)	2014	1/71,000	13	SCA+D	All	MN high school athletes	Public Media Reports (Parent Heart Watch)	14–18	2003–2012
Maron et al. (6)	2014	1/63,000	64	SCD	All	NCAA athletes	U.S. National Registry of Sudden Death in Athletes; NCAA Memorial Resolutions List	17–26	2002–2011
Harmon et al. (7)	2015	1/54,000	79	SCD	All	NCAA athletes	1) NCAA Resolutions List; 2) Parent Heart Watch database; and 3) NCAA insurance claims.	18–26	2003–2013
Harmon et al. (12)	2016	1/101,000 SCD 1/67,000 SCD+SCA	69 SCD 104 SCA+SCD	SCD SCA+SCD	All	High school athletes from 7 states (CA, FL, MN, NJ, OH, TN, and TX)	Parent Heart Watch database	14–18	2007–2013

C/T = competition/training (defined as during or within 1 h of cessation of activity); EMS = emergency medical service; NCAA = National Collegiate Athletic Association; NCCSIR = National Center for Catastrophic Sports Injury Research; NHLBI = National Heart, Lung, Blood Institute; PE = physical education; SCA = sudden cardiac arrest; SCD = sudden cardiac death.

CAUSES OF SUDDEN CARDIAC DEATH

The cardiac abnormalities contributing to SCD in athletes ≤ 35 years of age generally fall into 3 categories: electrical, acquired, and structural cardiac abnormalities (Figure 1). Most of these abnormalities are inherited cardiac disorders that may be quiescent

but can predispose the athlete to SCD primarily through ventricular arrhythmia (13). Primary, inherited electrical cardiac abnormalities were implicated in $\leq 5\%$ of confirmed SCD events in the U.S. Registry (8) and $<4\%$ in NCAA athletes (6,7), and 0% were identified in studies from the United Kingdom (36) and Australia and New Zealand (37). Acquired

TABLE 2 Comparison Between AHA 14 Points and the PPE-4 Elements

AHA Recommendations	Pre-Participation Physical Evaluation
Medical history* Personal History 1. Chest pain/discomfort/tightness/pressure related to exertion 2. Unexplained syncope/near syncope† 3. Excessive and unexplained dyspnea/fatigue or palpitations, associated with exercise 4. Prior recognition of a heart murmur 5. Elevated systemic blood pressure 6. Prior restriction from sports 7. Prior testing for heart disease, ordered by a physician	Heart Health Questions About You 6. Have you ever had discomfort, pain, tightness, or pressure in your chest during exercise? 5. Have you ever passed out or nearly passed out DURING or AFTER exercise? 12. Do you get more tired or short of breath more quickly than your friends during exercise? 10. Do you get lightheaded or feel more short of breath than expected during exercise? 7. Does your heart ever race or skip beats (irregular beats) during exercise? 8. Has a doctor ever told you that you have any heart problems? If so, check all that apply: ☐ High blood pressure ☐ A heart murmur ☐ High cholesterol ☐ A heart infection ☐ Kawasaki disease Other: _____ 1. Has a doctor ever denied or restricted your participation in sports for any reason? 9. Has a doctor ever ordered a test for your heart? (For example, ECG/EKG, echocardiogram) 11. Have you ever had an unexplained seizure?
Family History 8. Premature death (sudden and unexpected or otherwise) before 50 yrs of age attributable to heart disease ≥1 relative 9. Disability from heart disease in a close relative <50 yrs of age 10. Hypertrophic or dilated cardiomyopathy, long QT syndrome or other ion channelopathies, Marfan syndrome, or clinically significant arrhythmias; specific knowledge of genetic cardiac condition in family member	Heart Health Questions About Your Family 13. Has any family member or relative died of heart problems or had an unexpected death before age 50 (including drowning, unexplained car accident, or sudden infant death syndrome)? 14. Does anyone in your family have hypertrophic cardiomyopathy, Marfan syndrome, arrhythmogenic right ventricular cardiomyopathy, long QT syndrome, short QT syndrome, Brugada syndrome, or catecholaminergic polymorphic ventricular tachycardia? 15. Does anyone in your family have a heart problem, pacemaker, or implanted defibrillator? 16. Has anyone in your family had unexplained fainting, unexplained seizures, or near drowning?
Physical examination 11. Heart murmur‡ 12. Femoral pulses to exclude coarctation 13. Physical stigmata of Marfan syndrome 14. Brachial artery blood pressure (sitting position)§	Physical examination a. Heart ■ Murmurs (auscultation standing, supine, with or without Valsalva) ■ Location of point of maximal impulse b. Pulses ■ Simultaneous femoral and radial pulses c. Appearance ■ Marfan stigmata (kyphoscoliosis, high-arched palate, pectus excavatum, arachnodactyly, arm span > height, hyperlaxity, myopia, MVP, aortic insufficiency) d. Blood pressure
The table compares the 14-element American Heart Association recommendations for pre-participation cardiovascular screening of competitive athletes and the pre-participation physical evaluation monograph; differences are shown in bold. *Parental verification is recommended for high school and middle school athletes. †Judged not to be of neurocardiogenic (vasovagal) origin; of particular concern when occurring during or after physical exertion. ‡Refers to heart murmurs judged likely to be organic and unlikely to be innocent; auscultation should be performed with the patient in both the supine and standing positions (or with Valsalva maneuver), specifically to identify murmurs of dynamic left ventricular outflow tract obstruction. §Preferably taken in both arms. Reproduced with permission from Hainline et al. (78). AHA = American Heart Association; ECG = electrocardiogram; EKG = electrocardiogram; MVP = mitral valve prolapse; PPE-4 = Pre-Participation Physical Evaluation Monograph, 4th edition.	

abnormalities such as commotio cordis, which is increasingly reported in the United States and worldwide (38,39); myocarditis, which may have multiple causes (40); and environmental factors and performance-enhancing drugs have been linked with SCD in athletes (41,42). Anomalous coronary artery origins, particularly a coronary artery originating from the wrong aortic sinus, is a frequent cause of

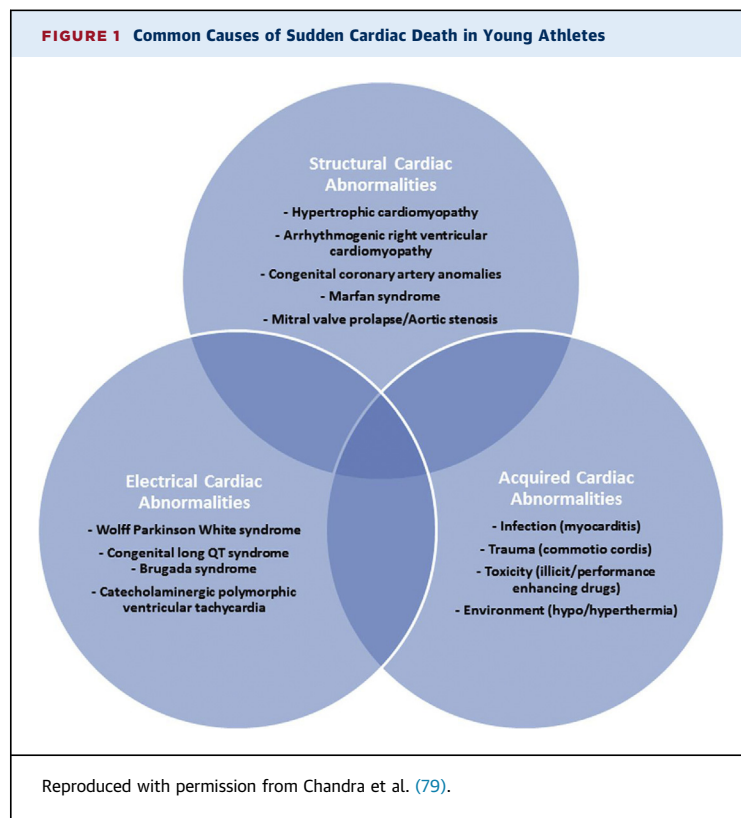
SCD, ranging from 7% to 17% of athlete cases (8,36,43). Atherosclerotic coronary artery disease is an uncommon but recognized cause of SCD in young athletes (8,13,37), whereas it becomes the dominant mechanism in older athletes (>60% of cases in athletes >35 years of age) (10,15,44-46).

Inherited structural cardiac conditions comprise most of the structural abnormalities resulting in SCD

in the young athletic population. Hypertrophic cardiomyopathy (HCM) is often cited as the most common cause of SCD in young competitive athletes, accounting for 36% of confirmed causes of SCDs in the U.S. National Registry of Sudden Death in Athletes (8,18), with a significant number of SCD cases labeled as “possible HCM” (8%). The prevalence of HCM in the general population is approximately 1:500 (0.2%) (47) but may in fact be less common (<0.07%) in athletes (48,49). More contemporary studies are starting to question the proportion of SCD attributed to HCM. A study from the United Kingdom found only 12% of SCDs in athletes was attributable to definitive HCM, with 25% of SCDs exhibiting idiopathic left ventricular hypertrophy (36). A prospective study of SCD among children and young adults conducted in Australia and New Zealand from 2010 through 2012 attributed only 6 of 54 cases (11%) of HCM during or post-exercise, whereas 20 were unexplained (37). Alpert et al. (50) condensed several published studies to generate an estimated risk of SCD for athletes due to HCM at 0.03% to 0.1%, which is comparable to contemporary published mortality rates in the overall HCM population (51).

Arrhythmogenic right ventricular cardiomyopathy (ARVC) is characterized by fibrofatty replacement of the RV myocardium with variable disease expression that can increase risk of SCD, notably during exertion (52). ARVC is the most common cause of SCD (24%) in athletes in the Italian registry from the Veneto region (13). It is also more common in the United Kingdom (10% among those ≤ 35 years of age) (36) than in the United States, including cohorts from the U.S. Registry of Sudden Death in Athletes, NCAA athletes, and U.S. military (6,8,43,53,54). The role that exercise may play in disease progression of individuals with genetic ARVC has evolved recently. In a study from James et al. (55), endurance athletes with desmosomal mutations developed symptoms at a younger age (30 vs. 41 years of age) and were more likely to develop ventricular tachycardia and ventricular fibrillation and heart failure. In a separate study from Sawant et al. (56), gene-elusive, nonfamilial ARVC was associated with more intense exercise prior to presentation and were more likely to be endurance athletes particularly among those presenting at <25 years of age. Unlike most other cases of cardiomyopathic processes, subjects who have possible or borderline cases of ARVC and probably even genotype-positive, phenotype-negative cases should not participate in competitive sports or even moderate to intense recreational exercise (Class III) (57).

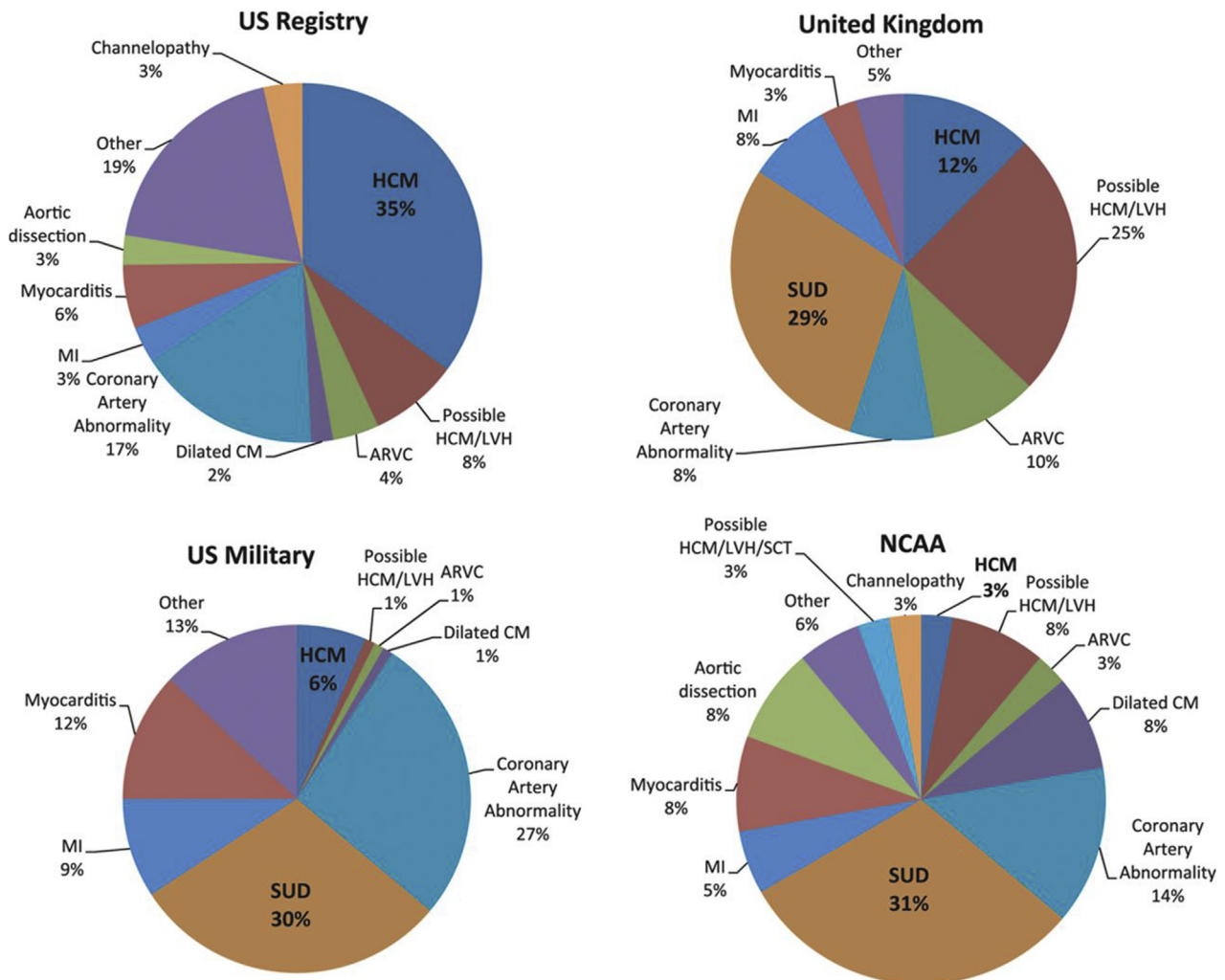
Dilated cardiomyopathy (DCM) is another potential cause of SCD in athletes. Mild forms of DCM can be



difficult to distinguish from “athlete’s heart.” Although a mildly reduced ejection fraction (EF; 40% to 50%) is not typical of “athlete’s heart,” it has been noted in several observational studies (58-60), particularly associated with increased left ventricular dimensions. These studies have also noted that the EF “augments normally” (59) or becomes “hyperdynamic” (60) with exercise, when observed by stress echocardiography. This adaptation likely reflects increased cardiac output, as attained primarily through increased stroke volume necessary to achieve the athletic prowess of a competitive athlete. As such, it would be anticipated that these subjects’ ejection fraction would increase normally with exercise (i.e., by stress echocardiography) and also be reflected in other objective measurements of fitness (i.e., $\text{V}_{\text{O}_2\text{max}}$) (61). Contemporary studies of myocardial contractile reserve as reflected by EF are needed to better classify the degree of augmentation by stress imaging that provides the best sensitivity and specificity for differentiating a mild DCM from the physiological adaptations of “athlete’s heart.”

Several studies have noted a higher than expected incidence of “morphologically normal hearts” than previously suspected (36,43,62). These cases are commonly referred to as sudden arrhythmic death syndrome (SADS), autopsy-negative SCD, or sudden

FIGURE 2 Comparison of Pathogenesis of Sudden Cardiac Deaths in Athletic Populations



ARVC = arrhythmogenic cardiomyopathy; CM = cardiomyopathy; HCM = hypertrophic cardiomyopathy; LVH = left ventricular hypertrophy; MI = myocardial infarction; NCAA = National Collegiate Athletic Association; SCT = sickle cell trait; SUD = sudden unexplained death. Reproduced with permission from Harmon et al. (43).

unexplained death. Studies from the United Kingdom (36,62), NCAA (7,43), and U.S. military (53,54) indicate that SADS may be the most common finding in post-mortem assessment (Figure 2). This group may in fact be comprised of a significant number of primary electrical disorders or channelopathies (63,64) that may be identified best by molecular autopsy. These are often difficult diagnoses to establish postmortem, unless specific protocols are followed to preserve material for genetic testing (64). In an ambitious study from Australia and New Zealand (37), all SCDs among children and young adults 1 to 35 years of age were prospectively studied from 2010 to 2012, with inherited cardiomyopathies representing 16% of the

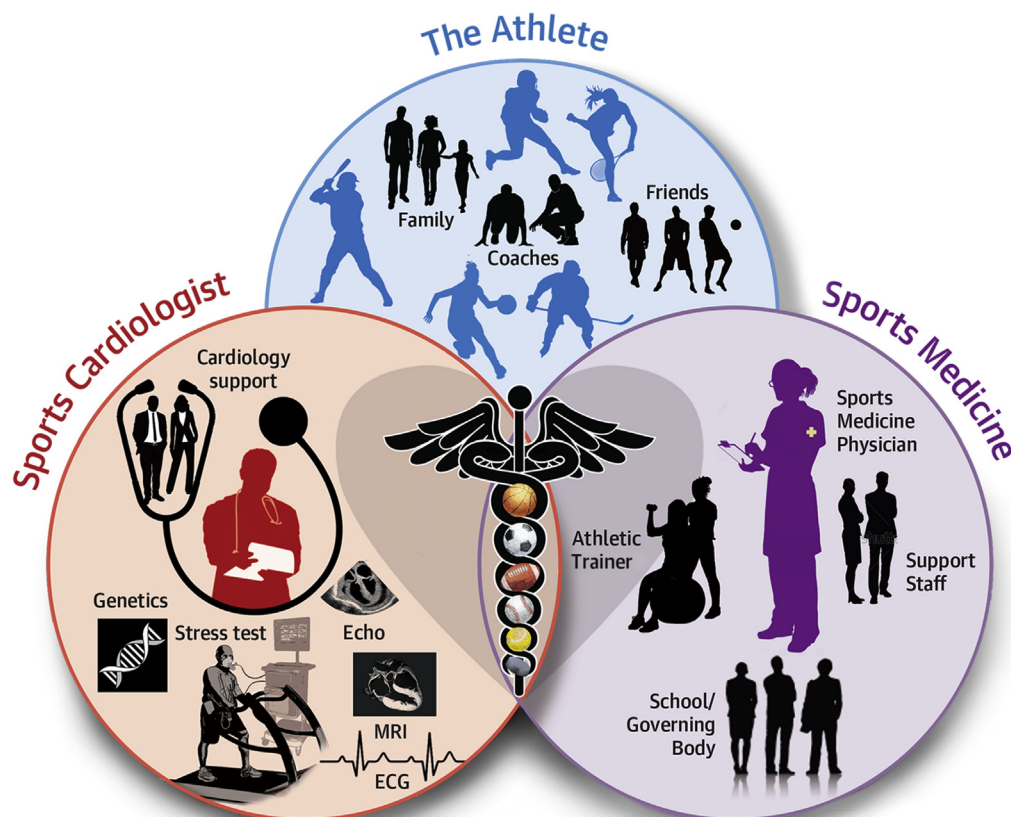
cases. Forty percent of the cases were unexplained at autopsy, with subsequent genetic analysis identifying pathogenic or probably pathogenic gene variants in 27%.

RISK STRATIFICATION AND RETURN TO PLAY

The individual with identified cardiac disease presents a challenge to the cardiac care team, the league/club, the athlete and their loved ones. The decision to disqualify or restrict an athlete is complex. The AHA/American College of Cardiology Foundation published scientific statements on eligibility and disqualification recommendations for

CENTRAL ILLUSTRATION Multidisciplinary Care of Athletes

Team-Based Approach to the Cardiovascular Care of Athletes



Emery, M.S. et al. *J Am Coll Cardiol HF*. 2018;6(1):30-40.

Multidisciplinary care of athletes focused on athlete-centered care ("Athlete Care Team") in the evaluation and management of athletes at risk of sudden cardiac death.

competitive athletes with cardiovascular abnormalities in 1985 (65), 1994 (66), 2005 (67), and 2015 (68). These documents inform the medical community as to consensus strategies to protect the athlete from harm while being mindful not to unnecessarily remove individuals from a healthy athletic lifestyle or competitive sports.

Although risk stratification for SCD exists for many cardiomyopathic processes within a general population setting, there is less to guide the clinician with regard to risk stratification of the

asymptomatic athlete who may have cardiac disease. This area of uncertainty is particularly evident in the 2015 AHA/ACC Eligibility and Disqualification Recommendations for Competitive Athletes with Cardiovascular Abnormalities, where a significant portion of the document(s) contains recommendations that carry Class II recommendations, where clinical and scientific uncertainty remain (68). Eligibility and disqualification criteria have been based almost exclusively on expert consensus, with limited data from a few, small observational registries primarily

with congenital long-QT syndromes and implantable defibrillators (69-73). This underscores the lack of studies in this unique population, particularly asymptomatic athletes, and the need for future research to better identify risk stratification paradigms that may help quantify the risks of participating in competitive athletics with certain cardiovascular disorders. Given these levels of uncertainty, participation in competitive athletics with a cardiovascular disorder should be a shared decision-making process (74) between the physician(s), the athlete, the family, and the school and/or governing body (Central Illustration).

EMERGENCY ACTION PLANS

Even with the best of primary and secondary prevention, SCD will occur, placing not only athletes but also event spectators at risk. The response to a sudden cardiac arrest at any athletic venue is an important component of any discussion of SCD in athletes. An athlete experiencing cardiac arrest can be effectively resuscitated with prompt recognition, rapid cardiopulmonary resuscitation, and application of an automated external defibrillator (AED). A study from Weisfeldt et al. (75) demonstrated that prompt application of an AED is associated with greater likelihood of survival (odds ratio: 1.75; 95% confidence interval: 1.23 to 2.50; $p < 0.002$), with the highest survival to hospital discharge noted in places of

recreation (49%) (75). An effective EAP necessitates the integration of adequate training (including coaches), rapid access to functioning AEDs, and effective communication with local EMS as well as the regular review and rehearsal of the emergency action plan (EAP) (76,77).

CONCLUSIONS

Sudden cardiac death in the athlete is a rare but catastrophic event for families, teams, leagues, and communities. The cardiac care team needs to be informed about strategies to identify those at risk and to respond to events. Although there is a significant body of knowledge addressing SCD in the athlete, much more needs to be done. Better epidemiology is needed to identify those at highest risk, and longitudinal registries should follow participants who have specific abnormalities. Observational or randomized trials need to be performed to test hypotheses regarding some of the alternative management strategies with clinical equipoise. It is the role of the professional to understand the issues, inform the athletes at risk, and protect those at highest risk.

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